Dhrubojyoti Gangopadhyay

Forward Deployed AI Engineer | Agentic Systems, MCP Integration & Enterprise Workflow Automation

Kolkata, India (open to remote and relocation) · +91 98740 76688 · dhrubojyoti.gangopadhyay@gmail.com
github.com/ganguydhrubo · linkedin.com/in/dhrubojyoti-gangopadhyay · sahkarmi.site (product) · atittle.com (product)

PROFESSIONAL SUMMARY

Applied AI engineer who ships. Sole builder of **Sahkarmi**, an open-source desktop AI-coworker platform (Electron, TypeScript, Claude Agent SDK, Model Context Protocol) with 50 tagged releases, a Microsoft Store listing and human approval gates on every sensitive action; also three live web products and a dozen public LangGraph and RAG prototypes. Preceded by 16 years of revenue-operations leadership at AGM level in cement, steel and industrial distribution, turning ambiguous dealer, pricing and rebate problems into audit-grade systems. I bridge executive stakeholders and code with no hand-off: discovery, scoping, build, deployment and adoption.

CORE COMPETENCIES

Agentic AI & LLM systems: LLM agents and tool use, Model Context Protocol (MCP) servers and clients, human-in-the-loop approval design, LangGraph multi-agent graphs, RAG (hybrid BM25 + vector, Chroma, pgvector), prompt engineering, LLM-as-judge evaluation

Engineering: TypeScript, Python, Node.js, Electron, SolidJS/React, Next.js, Astro, FastAPI, PostgreSQL/Supabase (RLS), Redis, Docker, Vitest, Playwright, pytest, GitHub Actions CI/CD, Cloudflare Workers/KV, Vercel, Render

Models & integrations: Anthropic Claude (Agent SDK), OpenAI Codex, Google Gemini, Groq, local models (OpenCode, Ollama); WhatsApp Cloud and Evolution API, Telegram Bot API, Composio, Razorpay, PayPal, Resend, Instagram Graph, X OAuth

Enterprise domain: Revenue operations, channel and distributor management, pricing, rebate and incentive systems, sales MIS, CRM/ERP workflows, AI adoption training and change management

PROFESSIONAL EXPERIENCE

Forward Deployed AI Engineer | 1XL Universe (Remote) **Jul 2026 – Present**

- Own AI adoption across HR, sales, marketing, operations, support, finance and technology; turn departmental problems into scoped LLM workflows and automations.
- Authored role-specific AI training curricula and field manuals (e.g. for executive assistants) and the company-wide induction programme for India and UAE offices.
- Advise on WhatsApp and agentic-workflow infrastructure (n8n, Evolution API) as the single technical-business bridge between engineering and business teams.

Founder & Applied AI Engineer | Dhrubo AI Studio, self-employed, Kolkata **Jan 2024 – Present**

- Built and shipped Sahkarmi (sahkarmi.site), a Windows desktop platform of AI coworkers with sandboxed browsers, scheduled routines, skills and MCP servers: 277 commits, 72 merged pull requests, 50 releases; listed on the Microsoft Store and Scoop.
- Shipped three live web products: atittle.com (Astro site with an AI website builder and one-click publishing to Cloudflare Workers KV), auraleads.online (multi-tenant lead-gen SaaS on Next.js/Supabase with Instagram, WhatsApp, Gmail and payment integrations) and zappy.rest (WhatsApp loyalty SaaS with Express API and Flutter client).
- Published 12+ public prototypes in Python and TypeScript: LangGraph multi-agent pipelines, hybrid RAG with evaluation harnesses, a Redis Streams fraud-scoring service and an EU AI Act compliance engine (green CI).

Revenue Operations Leadership | Cement, steel and industrial B2B distribution **2007 – Dec 2023**

- **AGM, Revenue Operations & Commercial Execution —** Shyam Steel / Tech Nirman Ispat (Nov 2022 – Dec 2023): worked inside dealer and distributor teams to turn fulfilment and collection problems into reporting frameworks that shortened executive decisions.
- **AGM, Revenue Operations & Channel Performance —** Star Cement (Feb 2021 – Nov 2022): grew market share across a multi-state North East India distributor network through pipeline visibility and execution discipline.
- **Senior Manager, Market Intelligence & Revenue Systems —** MP Birla Cement (Jun 2016 – Jan 2021): built audit-ready dealer rebate and incentive systems, correcting structural calculation errors and recovering leakage across a national network.
- Earlier channel and revenue roles at Lafarge India, ACC Limited, Ramco Cements and Johnson & Johnson (2007 – 2016).

SELECTED ENGINEERING PROJECTS

Sahkarmi — open-source desktop AI-coworker platform | *Electron 43, TypeScript, SolidJS, Claude Agent SDK, MCP, Vitest, GitHub Actions*

- Extended an Apache-2.0 base (OpenBot v0.1.11) by 110,000+ lines: multi-provider agent runtime (Claude Agent SDK with canUseTool gating, Codex app server, ACP bridge to OpenCode and Gemini), user-configurable MCP servers, Composio apps, Telegram control.
- Designed the approval architecture: 16 permission categories with non-overridable gates for payments, customer messages, publishing and shell access, time-boxed delegated authority, and a browser take-over flow so agents never handle passwords or OTPs.
- Hardened Electron (context isolation, sandbox, CSP, fuses, 209 sender-validated IPC channels); ~650 Vitest cases; tag-driven release pipeline with SHA-256 checksums, SPDX SBOM, build-provenance attestation and auto-update.

ATITTLE — AI website builder and publish pipeline | *Astro 7, Preact, TypeScript, Cloudflare Workers KV, Groq/Gemini/OpenRouter, Playwright*

- Multi-provider OpenAI-compatible LLM chain with health tracking and quotas; server-rendered sites written to Workers KV via the Cloudflare REST API and served by a custom Worker with strict CSP and custom-domain mapping; 109 Vitest and 70 Playwright tests.

LangGraph and RAG prototypes | *Python, FastAPI, LangGraph, LlamaIndex, ChromaDB, pgvector, Docker Compose*

- SENTRA-AI: 8-node planner, specialist and manager SOC-copilot graph with Anthropic, OpenAI-compatible, Azure and Gemini adapters behind JWT/RBAC. B2B Sales RAG: hybrid dense + BM25 retrieval with reciprocal-rank fusion, cross-encoder re-ranking and a 20-question LLM-judged eval harness (50 unit tests).

EDUCATION

MBA, Operations Management & Marketing | University of Kalyani, West Bengal, India **2005 – 2007**